

Runhua Wu

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RESEARCH INTERESTS

Queueing Systems, Stochastic Process.

EDUCATION

Georgia Institute of Technology

Atlanta, Georgia, U.S.A.

PhD in Operation Research

August 2023-Present

- Advisor: Hayriye Ayhan
- GPA: 3.85/4.0
- Passed Comprehensive Exams

Master in Statistics

August 2022-May 2023

- GPA: 4.0/4.0

Fudan University

Shanghai, China

B.Math. Minor in Computational Mathematics

September 2018-June 2022

- Scholarship: Math Undergraduate Scholarship

RESEARCH

Advisor: Hayriye Ayhan

- Investigated a queueing system with different types of impatient customers, completely characterized the optimal admission control policy.
- Investigated service control problem for systems with impatient customers, completely characterized the optimal service rate policy in queues with finite/infinite buffer space, and developed efficient algorithms to find the optimal policy
- Investigating server scheduling in queues with impatient customers, exploring the optimal non-anticipating scheduling policy in an M/G/1+M queueing system

PUBLICATIONS

Wu, R. and Ayhan, H., "Optimal Admission Control in Queues with Multiple Customer Classes and Abandonments", *Queueing Syst* 109, 6(2025).

Wu, R. and Ayhan, H., "Service Rate Control in Queues with Abandonments" (under major revision)

Wu, R., Ayhan, H., Down, D. G., "Maximizing Throughput in an M/G/1 Queue with Customer Abandonments" (submitted)

TALKS

"Optimal Admission Control in Queues with Multiple Customer Classes and Abandonments" at 2024 INFORMS Annual Meeting

October 2024

"Service Rate Control in Queues with Abandonments" at 2025 INFORMS Applied Probability Conference

July 2025

"Service Rate Control in Queues with Abandonments" at 2025 INFORMS Annual Meeting

October 2025

TEACHING

Teaching Assistant

Georgia Institute of Technology

- Undergraduate courses:

Regression and Forecasting (ISyE4031) May 2024-June 2024

Stochastic Manufacturing and Service Systems (ISyE3232) May 2025-August 2025

Holding office hours, grading homework assignments.

- Graduate courses:

Design and Analysis of Experiments (ISyE6413)

September 2024-December 2024

Stochastic Processes II (ISyE6762) January 2025-May 2025

Stochastic Processes I (ISyE6761) September 2025-December 2025

September 2026-Now

Computational Statistics (ISyE6416) January 2026-May 2026

Holding office hours, grading assignments and exams, tidying up notes.

COURSES TAKEN

13 PhD-level courses at GeorgiaTech:

Stochastic Processes I

Stochastic Processes II

Stochastic Calculus I

Stochastic Calculus II

Topics in Stochastics

Stochastic Optimization

High-Dimensional Probability

Linear Optimization

Probability I

Probability II

Queueing Theory

Simulation Theory

Nonlinear Optimization

Discrete Optimization

TECHNICAL STRENGTHS

Language Proficiency: Native Chinese Speaker, Proficient in English

Programming Languages: Python, Matlab, R, $\LaTeX$